Active-layer thickness prediction and regional differences across permafrost regions using a physics-residual machine-learning workflow

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Abstract

Active-layer thickness (ALT) varies from year to year, yet gridded products are mostly trained on multi-year conditions and rarely evaluated against same-location, same-year observations. We combined 19,172 cell-year observations from Alaska, the Lena Delta and north-western Canada (2010–2024) with ERA5-Land thawing degree-days accumulated to the observation day, and tested a physics-residual workflow in which gradient boosting learns the residual of a locally recalibrated Stefan model. Under spatial-block cross-validation, the physics-residual model reduced the Stefan error by 3.6 cm in the Lena Delta and 3.7 cm in Canada and matched it in Alaska. When the Lena Delta was withheld from training, direct machine learning exceeded the Stefan error by 18.0 cm, whereas the physics-residual model remained close to it. Twenty-five local labels then reduced its error from 23.8 to 19.9 cm, and five labels reduced the error at external CALM sites from 56.2 to 40.3 cm. In the same cells and years, the Lena Delta error (21.1 cm) was lower than that of five published products (28.4–34.6 cm). Anchoring the learner to the Stefan model therefore matters most where local labels are few or absent. The workflow provides year-labelled 1 km ALT maps for 2011–2024 with an area of applicability.

Keywords: permafrost, active-layer thickness, Stefan model, physics-guided machine learning, spatial cross-validation, ERA5-Land

Introduction

The active layer is the near-surface ground above permafrost that thaws each summer and refreezes each winter. Its thickness (ALT) sets the depth of seasonal thaw and controls ground stability, near-surface hydrology and the amount of soil organic carbon exposed to decomposition[1–3]. ALT responds to the thermal regime of individual years, so that interannual variability is an intrinsic part of the signal recorded by long-term monitoring programmes such as the Circumpolar Active Layer Monitoring (CALM) network[4,5]. Observations, however, are sparse and spatially clustered: most measurements originate from a small number of intensive field campaigns and long-term grids, and many sites are visited only once[6,7].

Several gridded ALT products at about 1 km resolution now cover the northern permafrost region. Process-based estimates such as ESA CCI Permafrost solve the ground heat equation forced by remotely sensed land-surface temperature[8,9], whereas statistical and machine-learning (ML) products relate site observations to climate, terrain and soil predictors[10–13]. Three limitations restrict their use for year-specific applications. First, most products are trained on multi-year mean predictors or on site-years pooled across many years, so that a product value is not tied to the climate of a stated year. Second, products are evaluated with different site sets and validation schemes; reported errors range, for example, from a random ten-fold root-mean-square error (RMSE) of 21.6 cm[12] to a leave-one-site-out RMSE of 60.5 cm[13], and cannot be compared with each other. Third, random cross-validation of spatially clustered observations overstates the accuracy of ML models at new locations[14–17], which obscures how a model would perform in a region without observations.

Physically based models offer a complementary route. The Stefan solution relates thaw depth to the thawing degree-days (TDD) of the season through a single coefficient that depends on soil thermal properties and moisture[4,18,19]. It reproduces the seasonal progression of thaw and responds to the climate of each year, but a single coefficient per region leaves systematic residuals associated with soil organic matter and terrain. Physics-guided ML, in which a physical model carries the dominant response and a learner corrects what the physics omits, has improved accuracy and extrapolation in several environmental applications[20–22], and in Alaska the Stefan model retained its accuracy on held-out sites better than a random forest[23]. It remains unclear how such a combination performs when the climate input is matched to the observation year, how it transfers to a region without labels, how many local labels a new region requires, and how it compares with published products in the same cells and years.

Here we address these questions with a cell-year design, in which each ALT observation is a 1 km cell in a stated year and each climate input is computed from ERA5-Land for that year and accumulated to the observation day[24]. Using 19,172 cell-year observations from 2010–2024 in Alaska, the Lena Delta and north-western Canada, we compare the Stefan model, direct ML and a physics-residual workflow in which gradient boosting learns the residual of a recalibrated Stefan model. In contrast to previous ALT mapping studies, which mostly report random or site-based cross-validation on product-specific site sets[11–13], all methods and products are scored here against the same observations and under the same validation designs. The study

makes four contributions. (i) It removes the temporal mismatch between ALT labels and climate inputs and quantifies the effect of the year-specific input. (ii) It evaluates physics-residual learning against direct ML, physics-input ML and physics pseudo-labels under five validation designs, including the hold-out of an entire region. (iii) It quantifies sample efficiency and transferability with label-number experiments within and across regions and in external CALM regions. (iv) It compares models and six published products in the same cells and years, with masks for product training sites, and provides year-labelled 1 km maps for 2011–2024 with an area of applicability and prediction intervals.

Data and methods

Study regions and active-layer observations

We used three regions with dense, multi-year thaw-depth observations: Alaska, the Lena Delta and north-western Canada (Fig. 1, Table 1). Observations for Alaska and Canada were taken from the ABoVE soil moisture and active-layer thickness compilation, version 2[6], and from the CALM end-of-season grid values[5]; observations for the Lena Delta were taken from the ALLena thaw-depth data set[7]. ABoVE probe measurements were restricted to August–October and to $0 < \mathrm{ALT} < 300$ cm, because measurements made earlier in the season record thaw depth rather than the seasonal maximum. ALLena measurements were restricted to August–September values without the probe-limit flag, and the seasonal maximum was taken for each location and year. All observations were assigned to 1 km cells and averaged within each cell and year, retaining the mean day of observation; CALM values without a date were assigned day 244 (1 September). We kept the years 2010–2024, for which year-specific climate inputs are available. The resulting labels comprise 15,540 cell-years in Alaska (2010–2024), 2,436 in the Lena Delta (2010–2022) and 1,196 in Canada (2010–2019 and 2021–2024), distributed over 73, 16 and 26 blocks of 0.5° (Table 1). Sources, versions and licences are listed in Supplementary Table S1, the years used in each analysis in Supplementary Table S2 and the processing steps in Supplementary Methods S1.

Year-specific climate input

Monthly ERA5-Land fields for January 2010 to December 2024[24] (2 m air temperature, top-layer soil temperature and snow water equivalent) were extracted at the nearest land grid cell of each 1 km cell. For cell c and year y, the thawing degree-day sum is

$$\mathrm{TDD}_{c,y} = \sum_{m=1}^{12} \max(T_{c,y,m},\, 0)\, d_{y,m}, \tag{1}$$

where $T_{c,y,m}$ is the monthly mean air temperature and $d_{y,m}$ the number of days in month m. Because most observations were made before the end of the thaw season, the model input is the TDD accumulated to the day of observation t,

$$\mathrm{TDD}^{\mathrm{obs}}_{c,y} = C_{c,y,k-1} + \frac{t - t_{k-1}}{d_{y,k}} \left(C_{c,y,k} - C_{c,y,k-1}\right), \tag{2}$$

Table 1 Data and years used. Top: ALT labels by region (cell-years of 2010–2024 with year-specific climate input). Bottom: inputs and products, with the period available and the years used in this study. Blocks are the 0.5° cells used for spatial cross-validation and bootstrap; there are no Canadian labels in 2020.

Region	Cell-years	Cells	Blocks	Years	Cells ≥4 yr	Largest year
Alaska	15,540	13,253	73	2010–2024	203	2014 (51%)
Lena Delta	2,436	2,206	16	2010–2022	25	2016 (28%)
Canada (NW)	1,196	730	26	2010–2024	48	2022 (37%)
External CALM	805	72	49	2010–2024	–	–

Data	Resolution	Available	Used	Role
ERA5-Land monthly [24]	0.1°	1950–	2010–2024; mean 2011–2024	Climate input
Copernicus DEM [25]	30 m	static	–	Terrain
SoilGrids 2.0 [26]	250 m	static	–	Soil
MODIS MOD13Q1, MOD10A2 [27,28]	250–500 m	2000–	2010–2024	Sensitivity test
ESA CCI permafrost probability [8]	1 km	1997–2021	mean	Map domain
Wei et al. [13]	1 km	2000–2024	2010–2024	Comparison
Liu et al. [12]	1 km	2003–2020	2010–2020	Comparison
ESA CCI ALT v4 [8]	1 km	1997–2021	2010–2021	Comparison
ESA CCI ALT v5 [9]	1 km	1997–2023	2010–2023	Comparison
Yi and Kimball [29]	1 km	2001–2015	2010–2015 (Alaska)	Comparison
Aalto et al. [10]	30″	2000–2014	time-invariant	Comparison

where $C_{c,y,k}$ is the cumulative TDD to the end of month k, k is the month containing t and t_{k-1} is the last day of the preceding month. Seven further climate variables were computed for each cell and year: the mean annual air temperature, $\sqrt{\text{TDD}}$, the warmest and the coldest monthly mean temperature, the freezing degree-days of the preceding winter (October $y-1$ to April y), the mean top-layer soil temperature and the mean snow water equivalent of the hydrological year (October $y-1$ to September y) (Supplementary Table S3, Supplementary Methods S2). Two alternative inputs were evaluated on the same cell-years: the full-season TDD of the observation year, and the 2011–2024 mean of the yearly values (mean climate input), which corresponds to the multi-year inputs of existing products. Yearly maps use the full-season TDD of each year and therefore represent end-of-season ALT.

Static covariates

Six terrain variables (elevation, slope, the sine and cosine of aspect, topographic position index and roughness) were computed from the 30 m Copernicus DEM [25] in a 33×33 pixel window (about 1 km) around each cell. Nine soil variables (clay, sand, silt, bulk density, coarse fragments and pH at 5–15 cm, and soil organic carbon at 0–5, 5–15 and 15–30 cm) were sampled from SoilGrids 2.0 at 250 m [26]. The main

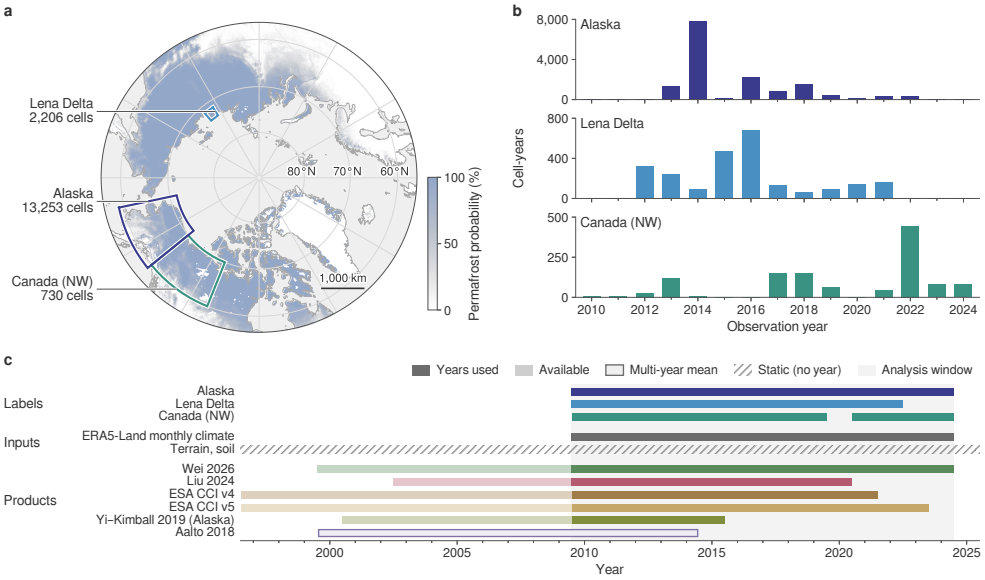


Fig. 1 Study regions, observations and data periods. a, ESA CCI permafrost probability (mean of 1997–2021) with the three study regions and the number of 1 km cells with ALT labels in 2010–2024. **b**, Number of cell-year labels per observation year, 2010–2024. **c**, Periods covered by the labels, the ERA5-Land monthly data used for the climate input, the static covariates and the six published products; for each product the period available and the years used for comparison at the label cells are distinguished, and the shaded band marks the analysis period 2010–2024.

feature vector $\mathbf{x}$ therefore contains 23 variables (eight climate, six terrain and nine soil). Sensitivity tests used nested sets: climate only; climate and terrain; the main set plus MODIS summer NDVI, EVI and snow-off timing for 2010–2024 (MOD13Q1 and MOD10A2, version 6.1)[27,28]; the main set plus high-resolution soil carbon and land-cover classes[30]; and all variables including the climate of the preceding year. The ESA CCI permafrost probability (mean of 1997–2021)[8] was used only to define the map domain (probability of at least 50%) and not as a predictor.

Model formulation

The Stefan model describes the thaw depth as

$$\widehat{\mathrm{ALT}}_{\mathrm{S}} = E\sqrt{\mathrm{TDD}^{\mathrm{obs}}}, \tag{3}$$

where E ($\mathrm{cm\,(^{\circ}C\,d)^{-1/2}}$) summarises the thermal properties and moisture of the soil[18,19]. The relation holds within a season at the Lena Delta monitoring sites and, with more scatter, among years at cells observed repeatedly (Supplementary Fig. S1). For a set of n training labels y_i with $s_i = \sqrt{\mathrm{TDD}_i^{\mathrm{obs}}}$, the coefficient is recalibrated by

least squares through the origin,

$$\hat{E} = \frac{\sum_{i=1}^{n} y_i \, s_i}{\sum_{i=1}^{n} s_i^2}. \tag{4}$$

The source-coefficient Stefan model uses $\hat{E}$ fitted on the other two regions, and the recalibrated Stefan model uses $\hat{E}$ fitted on the training labels of the target region. The physics-residual model retains the recalibrated Stefan prediction as an anchor and adds a learned correction (Fig. 2):

$$\widehat{\mathrm{ALT}} = \hat{E}\, s + \lambda\, g(\mathbf{x}), \qquad g = \arg\min_{f} \sum_{i=1}^{n} \left[\left(y_i - \hat{E}\, s_i \right) - f(\mathbf{x}_i) \right]^2, \tag{5}$$

where g is a CatBoost gradient-boosting regressor[31] (600 iterations, learning rate 0.03, depth 6, L2 regularisation 3) and $\lambda = 0.5$ shrinks the correction towards the physical anchor ($\lambda = 0.25$, 1 and a value selected by inner block cross-validation were tested as sensitivity cases). Three alternative structures place the physical information elsewhere: direct ML fits CatBoost to ALT from $\mathbf{x}$ alone; physics-input ML adds the recalibrated Stefan and Kudryavtsev[32] predictions as features; and physics pseudo-labels augment the training set with an equal number of rows drawn from the cells of the scoring fold, with their covariates and the training-fold Stefan prediction as the target. All learned models were fitted with three random seeds and their predictions averaged. Random forest, a multilayer perceptron, TabM and FT-Transformer[33–35] replaced CatBoost in a learner sensitivity test, and SHAP values[36] of the residual and direct models were computed for models fitted to all labels of each region.

Validation designs and uncertainty

All models were evaluated on out-of-fold predictions under five designs[14,15]: random five-fold cross-validation by cell-year; spatial-block five-fold cross-validation, with 0.5° blocks assigned to folds balanced by the number of cell-years (main design); year hold-out, in which each observation year with at least 20 cell-years is predicted by models trained on the other years; block-by-year hold-out, in which both the block fold and the year of the scored cell-years are excluded from training; and region hold-out, in which a region is predicted by models trained on the other two regions after removing training cells within 100 km of the test region. Accuracy is expressed as

$$\mathrm{RMSE} = \left[\frac{1}{n} \sum_{i=1}^{n} (\hat{y}_i - y_i)^2 \right]^{1/2}, \qquad \Delta\mathrm{RMSE}_{A,B} = \mathrm{RMSE}_A - \mathrm{RMSE}_B, \tag{6}$$

where both terms of ΔRMSE are computed on the same cell-years, so that negative values favour method A. Uncertainty was quantified with a paired block bootstrap: the 0.5° blocks were resampled with replacement $B = 2000$ times (1000 times for the label-number experiments), ΔRMSE was recomputed from all cell-years of the

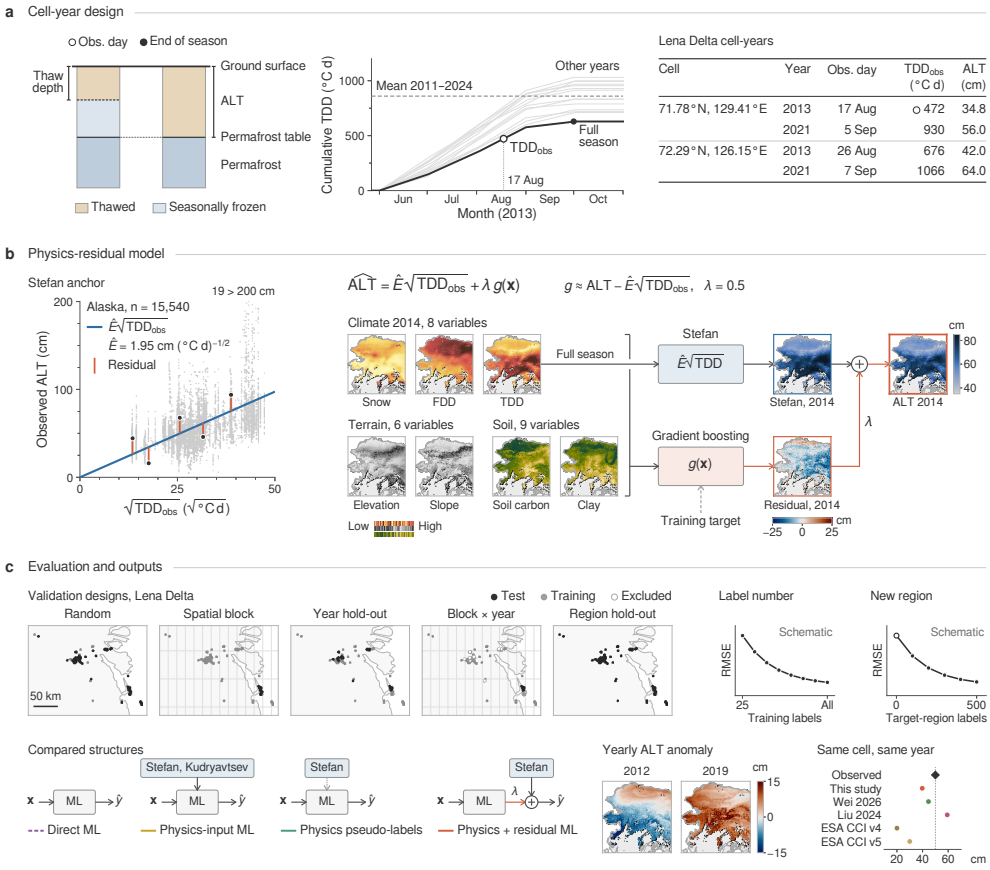


Fig. 2 Cell-year design, physics-residual model and evaluation. a, Each label is one 1 km cell in one year, and the climate input is the thawing degree-day sum accumulated to the observation day (TDD_{obs}), shown for a Lena Delta cell in 2013; the table lists cell-years of the years with the lowest (2013) and highest (2021) mean TDD of the Lena Delta label cells in 2011–2024. **b**, Stefan anchor fitted to the 15,540 Alaska cell-years and gradient boosting of its residual from 23 climate, terrain and soil inputs, added with the weight $\lambda = 0.5$; the maps use the full-season TDD of 2014 (Alaska). **c**, Validation designs on the Lena Delta label cells (one fold each; test year 2016), label-number and new-region experiments (schematic), compared model structures, yearly ALT anomalies relative to 2011–2024, and one Lena Delta cell-year (3 September 2013) compared with four products of the same year; this cell-year has the median absolute error of our model among the 42 cell-years with values of all four products and an observation after at least 90% of the seasonal TDD.

resampled blocks, and the 2.5th and 97.5th percentiles of the replicates define the 95% interval,

$$\mathrm{CI}_{95} = \left[Q_{0.025}\left(\Delta\mathrm{RMSE}^{(b)}\right),\ Q_{0.975}\left(\Delta\mathrm{RMSE}^{(b)}\right) \right], \quad b = 1, \ldots, B. \tag{7}$$

We describe a difference as resolved when its interval excludes zero. Conditions of the gain were examined within regional terciles of eight covariates (full-season TDD of

the observation year, freezing degree-days, snow water equivalent, soil organic carbon, clay, slope, elevation and summer NDVI); terciles with fewer than 30 cell-years or fewer than five blocks were not evaluated. Year-wise RMSE was computed for years with at least 30 cell-years, and the correlation between observed and predicted year-to-year changes was computed at cells observed in several years.

Label-number and transfer experiments

Within each region, the training labels of each spatial-block fold were subsampled to n = 25, 50, 100, 250, 500, 1000, 2500 or all (three random draws), the Stefan coefficient was recalibrated on the same n labels, and the complete test fold was scored. For transfer, each region was scored in its spatial-block folds by models trained on all labels of the other two regions plus n = 0, 25, 50, 100, 250 or 500 labels drawn from the training folds of the target region; $\hat{E}$ was refitted on the n target labels (for $n \geq 3$), and the residual learner was trained on the source rows, anchored with the source coefficient, together with the target rows, anchored with the refitted coefficient. The external test used CALM observations of 2010–2024[5] outside the three study regions and outside mountain permafrost (Switzerland and Kazakhstan). Censored values, values flagged as early, burnt, flooded or not reaching the permafrost table, and sites within 100 km of the study labels were removed, leaving 816 cell-years in 73 cells in eight regions (Supplementary Methods S3). Each external region was scored block by block (cell by cell for Greenland, which forms one block) by models trained on the three study regions plus n = 0, 5, 10, 25, 50 or all labels from the other blocks of the same region. One Svalbard borehole with a mean ALT of about 6 m, outside the training range, was excluded from scoring, which left 805 cell-years in 72 cells and 49 blocks. The runs are listed in Supplementary Table S4.

Comparison with published products

Six products were sampled at the label cells: Wei et al.[13] (1 km, 2000–2024), Liu et al.[12] (1 km, 2003–2020), ESA CCI Permafrost ALT v4 (1997–2021) and v5 (1997–2023)[8,9], Yi and Kimball[29] (Alaska, 2001–2015) and Aalto et al.[10] (one value per cell for the 2000–2014 baseline). For yearly products the value of the observation year was used (Supplementary Methods S4). Products and models were compared on common cell-years, at which Wei et al., Liu et al. and ESA CCI v4 all provide a value for the observation year (2010–2020). Wei et al. published the coordinates of their training sites, and a leakage mask removed label cells within 25 km of these sites. Liu et al. were trained on an earlier version of the ABoVE observations used here[12], so their Alaska and Canada values are not independent of our labels. To test whether the residual learner can also correct a product, the product value p of the observation year was recalibrated on the training fold,

$$\hat{p} = a + b\,p, \tag{8}$$

and the residual $y - \hat{p}$ was learned with the same covariates, learner, λ and folds as in equation (5) (Supplementary Methods S5).

Yearly mapping, area of applicability and prediction intervals

For Alaska and the Lena Delta, the recalibrated Stefan and physics-residual models were fitted to all cell-years of the region and applied to the 1 km cells with an ESA CCI permafrost probability of at least 50% (858,517 and 38,086 cells), using the full-season climate of each year from 2011 to 2024. The area of applicability followed Meyer and Pebesma[37]. With $\mathbf{z}$ the standardised 23-dimensional feature vector (with the 2011–2024 mean climate), the dissimilarity of map cell k is $d_k = \min_j \|\mathbf{z}_k - \mathbf{z}_j\|_2$ over all training rows j, and the cell is inside the area of applicability when

$$d_k \le \tau, \qquad \tau = Q_{0.95}\left\{ \min_{j:\, b(j)\neq b(i)} \|\mathbf{z}_i - \mathbf{z}_j\|_2 \;:\; i \in \text{training rows} \right\}, \tag{9}$$

where $b(\cdot)$ denotes the spatial block. The 90% prediction interval of each map value was taken from the 5th and 95th percentiles of the spatial-block out-of-fold residuals. For display, the 1 km values were interpolated bilinearly onto a 250 m hillshade grid.

Computation

Models were fitted with fixed random seeds on cloud nodes (Rescale; 96-core CPU nodes, and one NVIDIA A100 GPU node for the neural learners); data preparation, aggregation and figures were run on a local server (Supplementary Table S4).

Code availability

The code for data preparation, modelling, evaluation and figures is available at [repository and DOI]. Every number in the text is generated from the stored result tables by the scripts that produce the manuscript macros.

Results

Effect of the year-specific climate input

The observations are unevenly distributed in time (Fig. 1b, Table 1). In Alaska, 51% of the cell-years were observed in 2014, whereas the Lena Delta labels are spread over 13 years. Most measurements were made in August (median day of year 224 in Alaska, 234 in the Lena Delta and 237 in Canada), before the end of the thaw season, and the TDD accumulated to the observation day amounted to a median of 0.73, 0.81 and 0.89 of the full-season sum (equation (2)).

Replacing the mean climate input by the year-specific input reduced the spatial-block RMSE of the recalibrated Stefan model in the Lena Delta (ΔRMSE −1.4 [−2.6, −0.8] cm) and that of the physics-residual model in the Lena Delta (−1.6 [−2.4, −1.0] cm) and Canada (−1.5 [−2.0, −0.4] cm) (Fig. 3a). In Alaska the change was not resolved (Stefan +1.8 [−0.1, +3.9] cm), consistent with the concentration of the labels in one year, for which the year-specific input adds little information beyond the spatial pattern. The year-specific input also ties each prediction to the climate of a stated year, which is the prerequisite for the yearly maps presented below.

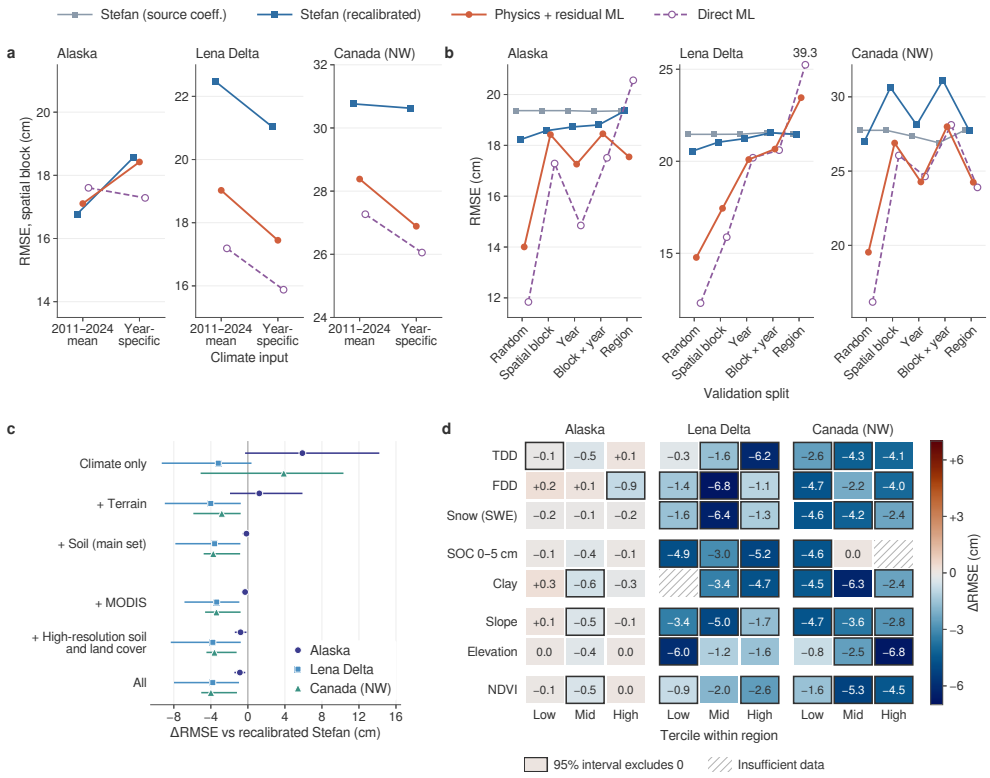


Fig. 3 Year-specific climate input, validation design and conditions of the residual gain. All panels use the cell-year labels of 2010–2024 and out-of-fold predictions. **a**, Spatial-block RMSE with the mean climate input (2011–2024 mean) and with the year-specific climate input (observation year, accumulated to the observation day). **b**, RMSE under five validation designs; under region hold-out the recalibrated Stefan model equals the source-coefficient model, and the value beyond the axis is printed. **c**, ΔRMSE between the physics-residual model and the recalibrated Stefan model for six covariate sets. **d**, The same ΔRMSE within regional terciles of eight covariates; outlined cells have 95% intervals excluding zero, and hatched cells contain fewer than 30 cell-years or five blocks. Intervals are 95% paired block-bootstrap intervals (2000 resamples of 0.5° blocks); negative values favour the physics-residual model.

Accuracy under spatial, temporal and regional validation

Model rankings depended on the validation design (Fig. 3b, Supplementary Table S5). Random cross-validation yielded the lowest errors for direct ML (11.8 cm in Alaska), and spatial-block cross-validation increased the direct-ML error by 5.4, 3.6 and 9.9 cm in Alaska, the Lena Delta and Canada. Year hold-out increased the direct-ML error in the Lena Delta from 15.9 to 20.2 cm, reflecting the spread of its labels over many years, and the block-by-year design produced the largest within-region errors for all methods fitted to labels of the region in all three regions. When an entire region was withheld, the direct-ML error in the Lena Delta rose to 39.3 cm, compared with 21.5 cm for the Stefan model with the coefficient of the other regions (ΔRMSE +17.8 [+8.0, +23.9] cm). In the same test the physics-residual model remained close to the

Stefan model (+2.0 [−0.8, +3.4] cm) and improved on it in Canada (−3.5 [−5.1, −0.1] cm). Spatial-block cross-validation is therefore used as the main design and region hold-out as the test of transferability.

Gain of physics-residual learning and its controls

Under spatial-block cross-validation with the year-specific input, the physics-residual model reduced the RMSE of the recalibrated Stefan model from 21.0 to 17.4 cm in the Lena Delta (ΔRMSE −3.6 [−7.8, −0.8] cm) and from 30.6 to 26.9 cm in Canada (−3.7 [−4.8, −0.8] cm), whereas both models reached about 18.4 cm in Alaska (−0.2 [−0.6, +0.1] cm). The gain depended on the covariates available to the residual learner (Fig. 3c). With climate covariates only, the residual model did not improve on the Stefan model in Alaska (+5.9 [−0.3, +14.2] cm) or Canada (+3.9 [−5.1, +10.3] cm); adding terrain and soil covariates produced the gains above. MODIS vegetation and snow timing (2010–2024) or high-resolution soil and land-cover classes changed the result by less than 0.5 cm in the Lena Delta and Canada, and the latter added a small gain in Alaska (−0.8 [−1.4, −0.2] cm).

The result depended on how the physical model entered the learner (Supplementary Fig. S2). Augmenting the training set with Stefan pseudo-labels reproduced the Stefan model (Lena Delta −0.3 [−0.4, −0.1] cm), and using the Stefan prediction as an input feature yielded results similar to direct ML. With all labels of a region, direct ML reached lower point errors than the physics-residual model (Alaska 17.3 versus 18.4, Lena Delta 15.9 versus 17.4 and Canada 26.1 versus 26.9 cm), but its difference from the recalibrated Stefan model was resolved only in Canada (−4.6 [−6.2, −0.2] cm), and its error increased under region hold-out. Replacing CatBoost by random forest, a multilayer perceptron, TabM or FT-Transformer did not change the sign of the physics-residual gain in the Lena Delta and Canada.

Across regional terciles of eight covariates, the gain occurred in most conditions of the Lena Delta (19 of 23 terciles with intervals excluding zero) and Canada (17 of 23) and increased with the error of the Stefan model; across terciles, the correlation between the Stefan error and the gain was −0.99 in the Lena Delta and −0.79 in Canada (Fig. 3d). In the warmest TDD tercile of the Lena Delta the gain reached −6.2 [−12.0, −0.7] cm, whereas no tercile in Alaska changed by more than 0.9 cm. These patterns indicate that the residual learner corrects departures of the effective Stefan coefficient from its regional value that arise from soil and terrain. Consistent with this interpretation, terrain variables accounted for 50% of the mean absolute SHAP value of the residual model in the Lena Delta and soil variables for 39% in Canada (Supplementary Fig. S3).

Sample efficiency and transfer to new regions

The label-number experiments show how the methods behave when labels are scarce (Fig. 4a). With 25 training labels drawn from the 2010–2024 cell-years of a region, the physics-residual model was at least as accurate as a Stefan model recalibrated on the same labels (Alaska −0.9 [−1.9, +0.1], Lena Delta −0.8 [−2.2, −0.1] and Canada −0.3 [−3.2, +1.7] cm), and with 100 labels its gain was resolved in all three regions (−0.7

[−1.4, −0.0], −2.0 [−5.3, −0.3] and −1.9 [−3.7, −0.3] cm). Direct ML trained on 25 labels had a higher point error than the Stefan model in Alaska (+2.8 [−0.3, +4.5] cm) and Canada (+1.2 [−1.5, +3.4] cm). Physics-input ML and physics pseudo-labels followed the patterns of their full-data results (Supplementary Fig. S4).

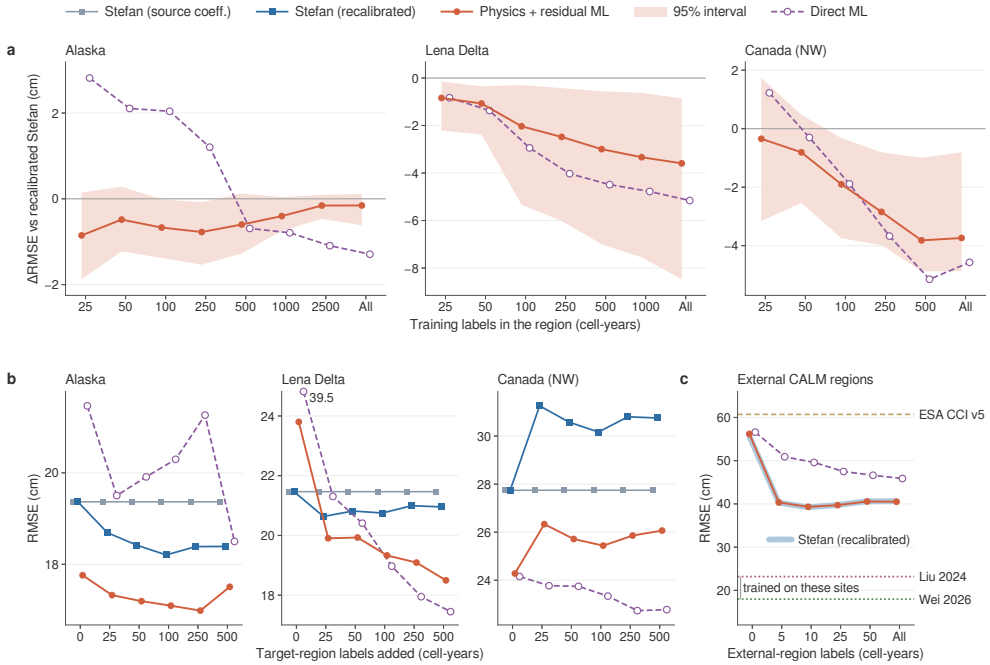


Fig. 4 Accuracy with few labels and in new regions. a, ΔRMSE between each method and a Stefan model recalibrated on the same n training labels within each region (labels of 2010–2024, three draws); the band is the 95% block-bootstrap interval of the physics-residual model. **b**, RMSE in each region for models trained on the other two regions plus n labels from the target region. **c**, RMSE in the external CALM regions (observations of 2010–2024) against the number of labels from the same external region, on the 474 cell-years with values of Wei et al., Liu et al. and ESA CCI v5 (2010–2020); horizontal lines mark the products, of which Wei et al. and Liu et al. include these sites in their training data.

In the transfer experiment, each region was predicted by models trained on the other two regions and n labels from the target region (Fig. 4b). Without target labels, direct ML reached 39.5 cm in the Lena Delta, compared with 21.5 cm for the source-coefficient Stefan model and 23.8 cm for the physics-residual model. Twenty-five target labels reduced the error of the physics-residual model to 19.9 cm and that of direct ML to 21.3 cm. In Alaska the physics-residual model gave the lowest error at every n (17.8 cm without target labels). In Canada, direct ML trained on the source regions was the most accurate (24.2 cm); recalibrating the Stefan coefficient on 25 Canadian labels increased the error from 27.7 to 31.3 cm, and the physics-residual model, which uses this anchor, followed (24.3 to 26.3 cm), reflecting the wide range of thaw depths among the Canadian sites.

The same procedure was applied to eight external CALM regions with 805 scored cell-years from 2010–2024 that were not used in model development (Fig. 4c, Supplementary Fig. S5). On the 474 cell-years of seven regions with values of Wei et al., Liu et al. and ESA CCI v5, the physics-residual model trained on the three study regions reached 56.2 cm without local labels. Five labels from the same external region reduced the error to 40.3 cm and 25 labels to 39.7 cm, whereas direct ML reached 47.5 cm with 25 labels. With five labels the physics-residual model was more accurate than ESA CCI v5, which is not trained on ALT observations (60.7 cm on the same 474 cell-years; ΔRMSE −19.5 [−31.3, −9.2] cm on all 700 cell-years with an ESA CCI v5 value). Wei et al. and Liu et al., which include these sites in their training data, reached 18.0 and 23.2 cm. Most of the reduction came from recalibrating the Stefan coefficient; the residual learner did not reduce the error further at these sites (+0.4 [−0.5, +1.2] cm with five labels on all 805 cell-years), where the active layers of West and Central Siberia and Svalbard exceed the range of the study regions. The pooled values combine different regional responses: with five labels the error decreased in West Siberia, Central Siberia, European Russia and Svalbard, but increased in Chukotka (8.8 to 14.6 cm) and Kamchatka (7.9 to 11.7 cm), where the error without local labels was below 10 cm, and in north-eastern Siberia (32.6 to 34.8 cm).

Comparison with published products in the same cells and years

On the common cell-years (2010–2020), the physics-residual model reached 21.1 cm in the Lena Delta, compared with 28.4 cm for ESA CCI v5, 30.6 cm for Aalto et al., 31.0 cm for ESA CCI v4, 33.0 cm for Liu et al. and 34.6 cm for Wei et al. (Fig. 5a, Supplementary Table S6). Excluding cells within 25 km of product training sites, the paired differences were −12.5 [−15.6, −5.8] cm relative to Wei et al. and −5.0 [−9.5, −2.5] cm relative to ESA CCI v5. In Alaska (17.1 versus 17.9 and 18.8 cm for Wei et al. and Liu et al.) and Canada (25.8 versus 27.0 and 27.9 cm), the model and the two ML products performed on par.

The regional contrast is consistent with the training data of the products. Alaska label cells lie at a median distance of 7 km from the training sites of Wei et al., and Liu et al. were trained on an earlier version of the same Alaska and Canada field data, whereas the Lena Delta cells lie at a median distance of 107 km (Supplementary Fig. S6). Both products overestimate the thin active layers of the northern delta, with mean biases of +21.1 and +19.9 cm north of 72°N. A model trained without any Lena Delta label was already more accurate there than Wei et al. on the same cell-years (26.0 versus 30.3 cm; ΔRMSE −4.3 [−6.1, −0.8] cm), indicating that the advantage in this region does not depend on local labels alone.

Maps of the same year show these differences in space (Fig. 6). For Alaska in 2014, the three maps agree at the observed cells (RMSE 17.1 cm for the out-of-fold predictions of our model, 15.8 cm for Wei et al. and 16.4 cm for Liu et al.), of which 85 of 95 lie within 25 km of a training site of Wei et al., and the largest differences between maps occur in mountain terrain above 600 m, which lies largely outside the area of applicability and where the products are deeper than our map by +21.9 and +55.2 cm between 68 and 70°N. For the Lena Delta in 2016, where all 14 observed

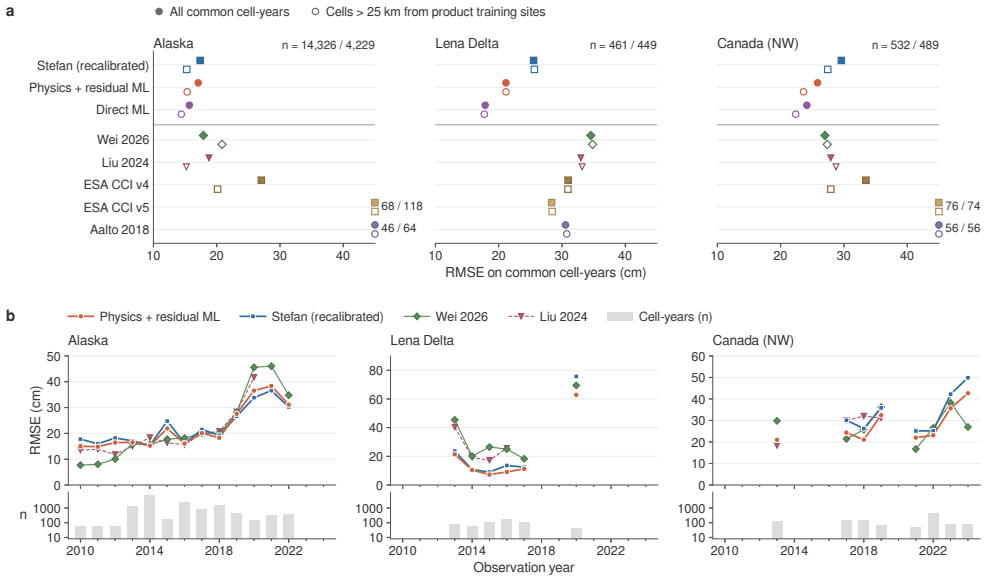


Fig. 5 Comparison with published products in the same cells and years. a, RMSE of three methods (spatial-block out-of-fold predictions) and five products on the common cell-years (2010–2020), for all cells and for cells more than 25 km from the training sites of Wei et al.; ESA CCI v5 and Aalto et al. are scored where they have a value; values beyond the axis are printed. **b**, RMSE per observation year on the same cell-years of each year, for years with at least 30 cell-years (Alaska 2010–2022; Lena Delta 2013–2017 and 2020; Canada 2013, 2017–2019 and 2021–2024); Liu et al. end in 2020; bars give the number of cell-years per year.

cells lie more than 25 km from these training sites, both products are deeper than the observations across the delta (mean bias +23.6 and +27.3 cm; RMSE 26.2 and 30.4 cm), whereas our model reaches an RMSE of 12.7 cm.

Year by year, the physics-residual model had a lower error than both Wei et al. and the Stefan model in all 6 Lena Delta years with at least 30 cell-years (2013–2017 and 2020), in 7 of 13 Alaska years against Wei et al. and 9 of 13 against the Stefan model (2010–2022), and in 7 of 8 Canadian years against the Stefan model (Fig. 5b). At Alaska cells observed in several years, the correlation between observed and predicted year-to-year changes was 0.28 for the physics-residual model and 0.15 and 0.12 for Wei et al. and Liu et al., whereas in the Lena Delta the products followed the interannual changes more closely (0.55 and 0.42 versus 0.29).

The residual learner can also be applied with a product as the anchor (equation (8)). With the Lena Delta labels, adding the residual learner to the recalibrated product of Wei et al. changed its error by −2.3 [−7.4, −0.4] cm, and the Stefan-anchored model remained more accurate than the product-anchored model (−2.8 [−6.2, −0.1] cm; Supplementary Fig. S7). In Alaska and Canada the two anchors were not distinguishable.

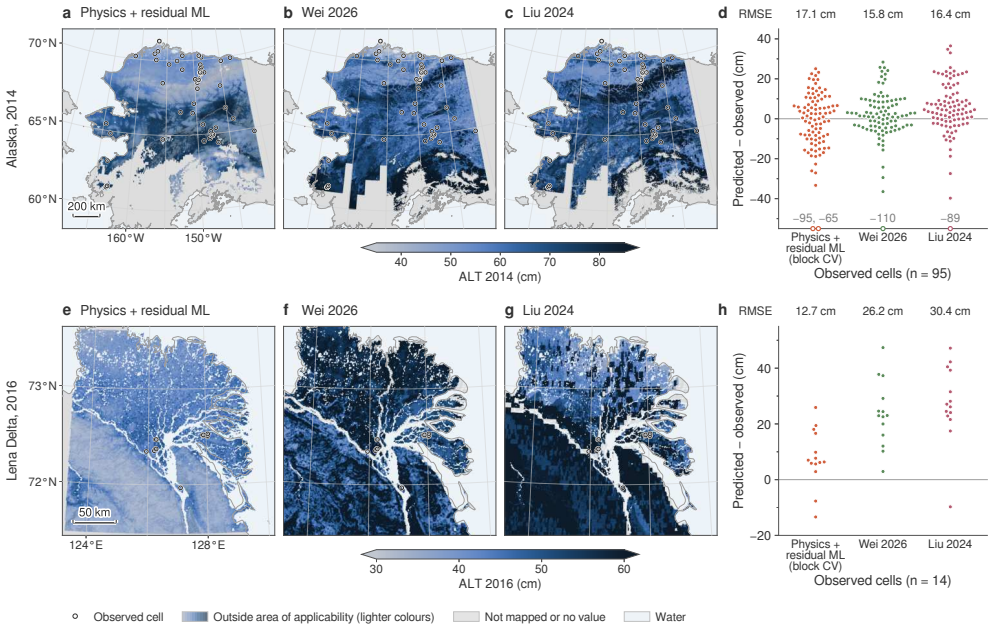


Fig. 6 ALT maps of the physics-residual model and two published products for the same year. a–c, Alaska in 2014: physics-residual model, Wei et al. and Liu et al. on a common colour range. **d**, Predicted minus observed ALT at the 95 cells observed in 2014 with values in all three maps, and the RMSE of each map; for our model the predictions are spatial-block out-of-fold values. Values beyond the axis are printed. Product values are not masked for training sites; 85 of the 95 cells lie within 25 km of a training site of Wei et al. **e–h**, The same for the Lena Delta in 2016 (14 cells, all more than 25 km from these training sites). The map domain is the area with an ESA CCI permafrost probability of at least 50%; the 1 km values are interpolated to 250 m for display.

Year-labelled 1 km ALT maps, 2011–2024

The physics-residual models fitted to all cell-years of Alaska and the Lena Delta were applied with the ERA5-Land climate of each year from 2011 to 2024 (Fig. 7). The maps cover 858,517 cells in Alaska and 38,086 in the Lena Delta. Relative to the 2011–2024 mean, the regional mean anomaly ranged from −3.2 cm in 2012 to +7.1 cm in 2019 in Alaska and from −9.4 cm in 2017 to +10.0 cm in 2020 in the Lena Delta, with spatial patterns that differ between years (Fig. 7b). In the Lena Delta, the mean anomaly of the observed cells had the same sign as the predicted regional anomaly in 8 of 10 years with at least 50 cell-years (2012–2021; $r = 0.75$), and the correlation remained 0.72 without 2020, when the observed anomaly of 143 cell-years reached +46.9 cm. In Alaska, where the set of observed cells changes between campaigns, the agreement was weaker ($r = 0.25$), and the predicted interannual range (10.2 cm) was smaller than that of the observed cell means (17.7 cm) (Fig. 7c). The area of applicability covers 56% of the Alaska cells and 62% of the Lena Delta cells; cells outside it lie mainly in mountains and at the southern margin of the permafrost zone (Supplementary Fig. S8). The half-width of the 90% prediction interval is 25.5 and 24.5 cm. The correction added by the residual learner is mostly within ±10 cm

(Supplementary Fig. S9), and series at long-record cells are shown in Supplementary Fig. S10.

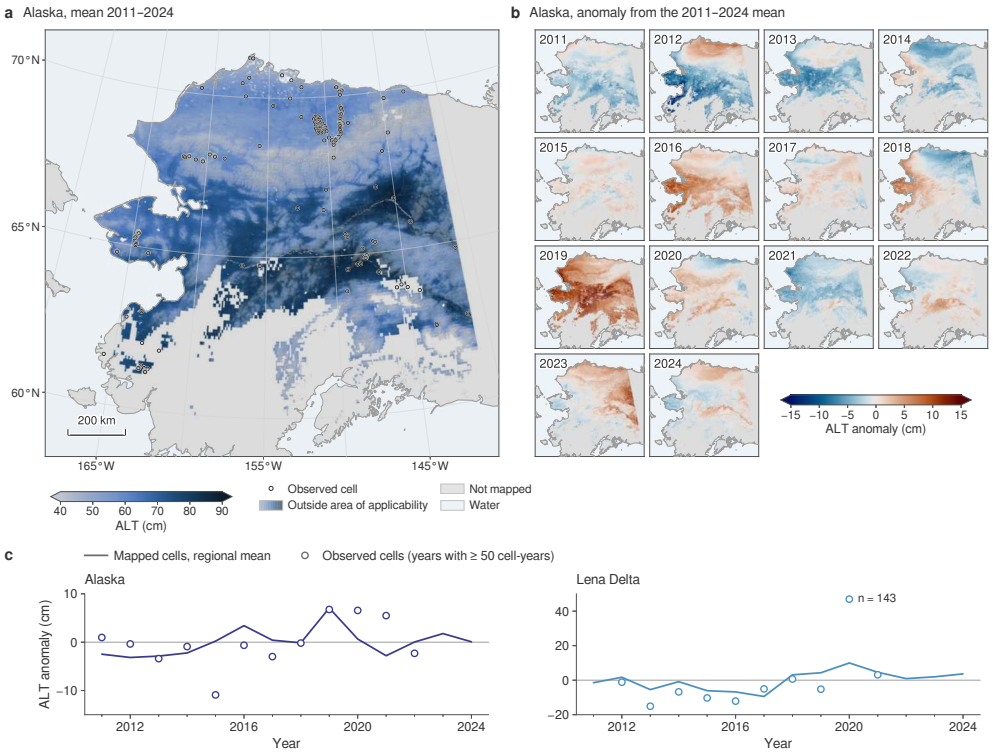


Fig. 7 Year-labelled 1 km ALT maps, 2011–2024. a, Mean ALT of 2011–2024 from the physics-residual model in Alaska, computed from the yearly maps; each yearly map uses the full-season ERA5-Land climate of that year; lighter colours mark cells outside the area of applicability, and grey land lies outside the map domain. **b**, ALT anomaly of each year relative to the 2011–2024 mean, on a common colour range (±15 cm). **c**, Regional mean anomaly of the predicted maps and mean anomaly of the observed cells in years with at least 50 cell-years (Alaska 2011–2022, Lena Delta 2012–2021); observed anomalies are taken relative to the 2011–2024 map mean of each observed cell and centred on the mean predicted anomaly of the same years, and the Lena Delta 2020 value is labelled with its number of cell-years. The 1 km values are interpolated to 250 m for display.

Discussion

Physical anchoring and transferability

The results show that the way physical knowledge enters the model determines whether ML improves ALT prediction beyond a regional Stefan model and whether the improvement transfers to new regions. In the physics-residual formulation (equation (5)), the Stefan term carries the response to the thaw-season climate of each year, which extrapolates to new regions and years in a physically bounded form,

and the learner corrects departures of the effective coefficient associated with soil and terrain. Four results support this division of roles: the gain required terrain and soil covariates; it was largest where the Stefan error was largest; the attribution of the residual learner pointed to terrain in the Lena Delta and to soil organic carbon in Canada; and Stefan pseudo-labels pulled the learner towards the Stefan model without adding accuracy. Direct ML, which must learn the climate response from the labels, reached lower point errors than the physics-residual model when all labels of a region were available, but its Lena Delta error exceeded that of the Stefan model by 17.8 cm when the region was withheld. This contrast agrees with studies of physics-guided learning in limnology and hydrology, in which process knowledge mainly improves performance under data scarcity and extrapolation[20–22]. Because most permafrost regions have few or no ALT observations, the accuracy that a method retains without target-region labels is more relevant for mapping than its accuracy with all labels of a well-observed region.

Comparison with previous studies

Previous ALT mapping studies differ from the present one in the climate input, the validation design and the reference used for comparison. ML products relate observations pooled over many years to multi-year predictors and report random or site-based cross-validation on their own site sets[11–13]. In our data, random cross-validation underestimated the spatial-block error of direct ML by 3.6–9.9 cm, and withholding the Lena Delta increased its error there to 39.3 cm, compared with 21.5 cm for the Stefan model with the coefficient of the other regions. Errors from random or site-based splits therefore do not indicate the accuracy of an ML product in a region without training observations. Gautam et al.[23] compared a random forest with the Stefan model at 68 CALM sites in Alaska observed in 2014; the coefficient of determination of the random forest fell from 0.84 in training to 0.24 on a random test split, whereas the Stefan model retained 0.54. The present study extends this comparison to cell-years of 2010–2024 in three regions and to region hold-out, and combines the two approaches so that the learner corrects the Stefan model instead of replacing it. Process-based products such as ESA CCI Permafrost are not trained on ALT observations[8,9]; at the external CALM sites, the error of ESA CCI v5 (60.7 cm) exceeded that of our model with five local labels (40.3 cm).

Scoring all models and products on the same cells and years, with masks for product training sites, makes these differences comparable. The regional ranking of the products followed their training data. Where the products were trained on nearby observations (Alaska and Canada), they were as accurate as our model; where they were not (the Lena Delta), they overestimated thin active layers and were less accurate than a model trained without any Lena Delta label. The residual learner also improved a product used as the anchor (equation (8)), so that local labels combined with residual learning can adapt both a physical model and an existing product to a region that the product did not learn. The same-cell, same-year protocol and the training-site masks can be applied to other products and regions.

Workflow for new regions and years

The label-number experiments translate into a sequence for applying the method to a new region. Without local labels, the Stefan model with the coefficient of other regions, or the physics-residual model trained on other regions, is a safer starting point than direct ML, whose Lena Delta error without target labels was 39.5 cm. Twenty-five Lena Delta labels reduced the error of the physics-residual model from 23.8 to 19.9 cm, below that of the source-coefficient Stefan model (21.5 cm), and in the external CALM regions five labels reduced the error from 56.2 to 40.3 cm. These results agree with the value of a limited number of local observations reported for regionalisation in hydrology[38]. The number of labels required depends on the distance of the new region from the training range: in Canada, where thaw depths vary widely, recalibration on 25 labels increased the error, and 25–100 labels that span the local range of soils and terrain are a reasonable target. Because each map is driven by the ERA5-Land climate of a stated year and carries an area of applicability and a prediction interval, the same workflow produces yearly maps for monitoring, allows comparison with observations or products of the same year, and marks the areas outside the area of applicability, where additional observations are needed.

Limitations and outlook

Half of the Alaska labels come from 2014, which prevented a resolved estimate of the effect of the year-specific input in that region. The Lena Delta labels come from one data set in 16 blocks, which widens the confidence intervals. The climate input is monthly and the soil covariates are static, so that short-term weather and changes in soil moisture are not represented. Active layers thicker than about 3 m, mainly in mountains and at the southern margin of the permafrost zone, lie outside the training range, and the external tests show that the residual learner does not transfer to such conditions without local labels. Daily climate inputs, information on snow and soil moisture, and labels from a fourth region such as West Siberia, where CALM and campaign data both exist, are the next steps.

Conclusions

We evaluated active-layer thickness models in a cell-year design, in which labels, climate inputs and scores refer to the same 1 km cell and year. Matching the climate input to the observation year reduced the spatial-block error where the observations span many years and assigns every prediction to a stated year. A Stefan model recalibrated with local labels and corrected by a gradient-boosting model of its residual reduced the Stefan error by 3.6 cm in the Lena Delta and 3.7 cm in Canada, remained close to the Stefan model in a region withheld from training, and reduced its error in the Lena Delta and the external CALM regions with 5–25 local labels. Direct ML reached lower point errors within regions with all labels, but without Lena Delta labels its error there exceeded that of the Stefan model by 18.0 cm. In the same cells and years, the model was more accurate than five published products in a region that the products did not learn and performed on par with Wei et al. and Liu et al. where

these products were trained on nearby observations. Physical anchoring of the learner is therefore most useful where target-region labels are few or absent, which is the common situation in permafrost regions. The workflow provides year-labelled 1 km ALT maps with an area of applicability and prediction intervals, together with a protocol for evaluating and adapting ALT models and products in other permafrost regions.

Data availability. All input data are publicly available. ALT observations: ABoVE soil moisture and active-layer thickness, version 2 (ORNL DAAC, https://doi.org/10.3334/ORNLDAAC/2369), ALLena thaw-depth data (PANGAEA, https://doi.org/10.1594/PANGAEA.973813) and GTN-P CALM (PANGAEA, https://doi.org/10.1594/PANGAEA.972777). Climate: ERA5-Land monthly averaged data (Copernicus Climate Data Store). Covariates: Copernicus DEM GLO-30, SoilGrids 2.0, ESA CCI permafrost probability, and MODIS MOD13Q1 and MOD10A2 version 6.1 (accessed through AppEEARS). Products: Wei et al. (https://doi.org/10.5281/zenodo.21667583), Liu et al. (https://doi.org/10.5281/zenodo.10070610), ESA CCI ALT v4 and v5 (CEDA), Aalto et al. (https://zenodo.org/records/5003804) and Yi and Kimball (https://doi.org/10.3334/ORNLDAAC/1760). The cell-year label table, the yearly climate-input table, the static covariate table, the out-of-fold predictions of all methods and the yearly 1 km maps of Alaska and the Lena Delta (2011–2024) will be deposited in a public repository upon acceptance [repository and DOI].

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Author contributions. S.B. designed the study, compiled the data, performed the analyses and wrote the manuscript. [Co-author contributions.]

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Competing interests The authors declare no competing interests.